Lecture 11

Universal enveloping algebras and PBW

Objectives. *A representation turns brackets into commutators of operators. We build the associative algebra that does this universally. Then we prove that its ordered monomials form a basis.*

By the end of this lecture you will be able to:

- Construct $U(\mathfrak{g})$ and pass between $\mathfrak{g}$-modules and $U(\mathfrak{g})$-modules.
- Prove the Poincaré–Birkhoff–Witt theorem, including the step that uses the Jacobi identity.
- Identify $\operatorname{gr} U(\mathfrak{g})$ with $\operatorname{Sym}(\mathfrak{g})$ and factor $U(\mathfrak{g})$ along a triangular decomposition.
- Reorder words in $U(\mathfrak{sl}_2)$ and compare the result in the associated graded algebra.

Throughout, $\mathfrak{g}$ is a finite-dimensional complex Lie algebra. An *algebra* is an associative complex algebra with 1, and algebra homomorphisms preserve 1. This lecture departs from the standing convention on dimension: tensor algebras, polynomial algebras, $U(\mathfrak{g})$ and their modules are infinite-dimensional in general. All arguments are algebraic.

11.1 The enveloping algebra

For a vector space V, the *tensor algebra* is

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \qquad V^{\otimes 0} = \mathbb{C},$$

with product given on pure tensors by concatenation. Multilinear algebra gives its universal property. A linear map $\varphi : V \to A$ into an algebra extends uniquely to an algebra homomorphism $T(V) \to A$, given by

$$v_1 \otimes \cdots \otimes v_k \longmapsto \varphi(v_1) \cdots \varphi(v_k).$$

Every algebra A is a Lie algebra under $[a, b] = ab - ba$. The Jacobi identity holds by the expansion in Lecture 1.

Definition 11.1 (Universal enveloping algebra)**.** Let I be the two-sided ideal of $T(\mathfrak{g})$ generated by the elements

$$x \otimes y - y \otimes x - [x, y] \qquad (x, y \in \mathfrak{g}).$$

The *universal enveloping algebra* is $U(\mathfrak{g}) = T(\mathfrak{g})/I$. Let $\iota : \mathfrak{g} \to U(\mathfrak{g})$ be the inclusion $\mathfrak{g} = \mathfrak{g}^{\otimes 1} \subseteq T(\mathfrak{g})$ followed by the quotient map.

By construction,

$$\iota(x)\iota(y) - \iota(y)\iota(x) = \iota([x,y]) \quad \text{for all } x, y \in \mathfrak{g}.$$

So ι is a Lie algebra homomorphism into $U(\mathfrak{g})$ with its commutator bracket. Injectivity of ι is not yet known. It will follow from PBW.

Proposition 11.2 (Universal property). *Let A be an algebra and $\varphi\colon \mathfrak{g} \to A$ a linear map with*

$$\varphi([x,y]) = \varphi(x)\varphi(y) - \varphi(y)\varphi(x) \quad \textit{for all } x, y \in \mathfrak{g}.$$

There is a unique algebra homomorphism $\tilde{\varphi}\colon U(\mathfrak{g}) \to A$ with $\tilde{\varphi} \circ \iota = \varphi$.

Proof. Let $\Phi\colon T(\mathfrak{g}) \to A$ extend φ. On a generator of I,

$$\Phi(x \otimes y - y \otimes x - [x,y]) = \varphi(x)\varphi(y) - \varphi(y)\varphi(x) - \varphi([x,y]) = 0.$$

The kernel of Φ is a two-sided ideal, so it contains I. Hence Φ factors through $U(\mathfrak{g})$.

The algebra $U(\mathfrak{g})$ is generated by $\iota(\mathfrak{g})$, because $T(\mathfrak{g})$ is generated by $\mathfrak{g}$. Two homomorphisms that agree on $\iota(\mathfrak{g})$ are therefore equal. ∎

Proposition 11.3 (Modules). *Let V be a vector space of any dimension, and let $\mathfrak{gl}(V)$ denote* End(V) *with the commutator bracket. The assignments $\rho \mapsto \tilde{\rho}$ and $\pi \mapsto \pi \circ \iota$ are inverse bijections between representations $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ and unital $U(\mathfrak{g})$-module structures $\pi\colon U(\mathfrak{g}) \to \mathrm{End}(V)$. A linear map between two modules is a $\mathfrak{g}$-intertwiner exactly when it is a $U(\mathfrak{g})$-module map.*

Proof. A representation is a linear map $\mathfrak{g} \to \mathrm{End}(V)$ satisfying the hypothesis of Proposition 11.2. That proposition gives $\tilde{\rho}$ with $\tilde{\rho} \circ \iota = \rho$. Conversely, $\pi \circ \iota$ preserves brackets because ι and π do. Uniqueness in Proposition 11.2 gives $\widetilde{\pi \circ \iota} = \pi$.

Let $T\colon V \to W$ be a linear map between two modules. Put

$$C_T = \{u \in U(\mathfrak{g}) : T\pi_V(u) = \pi_W(u)T\}.$$

The set C_T is a linear subspace, since its defining condition is linear in u. It contains 1 and is closed under products. For $u, u' \in C_T$,

$$T\pi_V(uu') = T\pi_V(u)\pi_V(u') = \pi_W(u)T\pi_V(u') = \pi_W(u)\pi_W(u')T.$$

So C_T is a subalgebra. It contains $\iota(\mathfrak{g})$ exactly when T is a $\mathfrak{g}$-intertwiner. In that case $C_T = U(\mathfrak{g})$, since $\iota(\mathfrak{g})$ generates $U(\mathfrak{g})$. ∎

In calculations we write x for $\iota(x)$. Until Corollary 11.8 this is only notation.

11.2 The filtration and spanning

Let U_p be the image of $\bigoplus_{k \le p} \mathfrak{g}^{\otimes k}$ in $U(\mathfrak{g})$. It is spanned by products of at most p elements of $\mathfrak{g}$. Every generator of I has positive tensor degree. The sum $\bigoplus_{k \ge 1} \mathfrak{g}^{\otimes k}$ is a two-sided ideal, so it contains I, and $1 \notin I$. Put $U_{-1} = 0$. Then

$$\mathbb{C} = U_0 \subseteq U_1 \subseteq U_2 \subseteq \cdots, \qquad \bigcup_p U_p = U(\mathfrak{g}), \qquad U_p U_q \subseteq U_{p+q}.$$

Fix an ordered basis $(x_1, \dots, x_n)$ of $\mathfrak{g}$. A nondecreasing sequence $\Sigma = (i_1 \le \cdots \le i_p)$ has *length* p. Its *ordered monomial* is

$$x_\Sigma = x_{i_1} x_{i_2} \cdots x_{i_p} \in U(\mathfrak{g}),$$

with $x_\varnothing = 1$. For an index λ, write $\lambda \le \Sigma$ if $\lambda \le i_1$. This holds for every λ when $\Sigma = \varnothing$.

Lemma 11.4 (Reordering). *Let $y_1, \dots, y_k \in \mathfrak{g}$ and $\sigma \in S_k$. Then*

$$y_{\sigma(1)} \cdots y_{\sigma(k)} - y_1 \cdots y_k \in U_{k-1}.$$

Proof. Write σ as a product of adjacent transpositions. Consecutive words in the resulting chain differ by swapping two neighbours a, b. With u and v the products of the other $k-2$ factors,

$$u\,ab\,v - u\,ba\,v = u\,[a,b]\,v \in U_{k-1}.$$

Summing these differences along the chain proves the claim. ∎

Proposition 11.5 (Spanning). *The ordered monomials of length at most p span U_p. Hence the ordered monomials span $U(\mathfrak{g})$.*

Proof. Use induction on p. For $p = 0$, the space $U_0 = \mathbb{C}$ is spanned by $x_\varnothing = 1$. By multilinearity, U_p is spanned by U_{p-1} and the words $x_{j_1} \cdots x_{j_p}$. Let Σ be the indices $j_1, \dots, j_p$ sorted into nondecreasing order. By Lemma 11.4,

$$x_{j_1} \cdots x_{j_p} - x_\Sigma \in U_{p-1}.$$

The induction hypothesis covers U_{p-1}. ∎

Theorem 11.6 (Poincaré–Birkhoff–Witt). *The ordered monomials x_Σ form a basis of $U(\mathfrak{g})$.*

Spanning is proved. For independence we build a module on which the ordered monomials act independently.

11.3 Independence of ordered monomials

Let $S = \mathbb{C}[z_1, \dots, z_n]$, and let S_p be the polynomials of degree at most p, with $S_{-1} = 0$. For Σ as above put $z_\Sigma = z_{i_1} \cdots z_{i_p}$. The monomials z_Σ of length at most p form a basis of S_p.

Lemma 11.7. *There is a bilinear map $\mathfrak{g} \times S \to S$, written $(x, z) \mapsto x \cdot z$, such that for all indices λ, μ and all $p \ge 0$:*

(A) $x_\lambda \cdot z_\Sigma = z_\lambda z_\Sigma$ *whenever* $\lambda \le \Sigma$*;*

(B) $x_\lambda \cdot z - z_\lambda z \in S_p$ *for every* $z \in S_p$*;*

(C) $x_\lambda \cdot (x_\mu \cdot z) - x_\mu \cdot (x_\lambda \cdot z) = [x_\lambda, x_\mu] \cdot z$ *for every* $z \in S$.

By (B) and linearity, $x \cdot z \in S_{p+1}$ for $x \in \mathfrak{g}$ and $z \in S_p$. Condition (C) is linear in x_λ and x_μ. Once it holds on basis vectors, it holds for all pairs of elements of $\mathfrak{g}$.

Proof. We define the action on $\mathfrak{g} \times S_p$ by induction on p, each stage extending the previous one. Stage p must satisfy (A) and (B) for $z \in S_p$, and (C) for $z \in S_{p-1}$. Call these conditions (A_p), (B_p) and (C_p). By the remark above, (C_p) uses only the action on S_p.

Stage 0. Put $x_\lambda \cdot 1 = z_\lambda$. Then (A_0) and (B_0) hold, and (C_0) is empty.

Stage $p \ge 1$: definition. Let z_Σ have length p.

(i) If $\lambda \le \Sigma$, put $x_\lambda \cdot z_\Sigma = z_\lambda z_\Sigma$.

(ii) Otherwise $\Sigma = (\mu, T)$ with $\mu < \lambda$, $\mu \le T$ and T of length $p-1$. By (B_{p-1}),

$$x_\lambda \cdot z_T = z_\lambda z_T + y, \qquad y \in S_{p-1}.$$

Put

$$x_\lambda \cdot z_\Sigma = z_\lambda z_\Sigma + x_\mu \cdot y + [x_\lambda, x_\mu] \cdot z_T.$$

The right side of (ii) uses only the action on S_{p-1}.

(A_p) and (B_p). Rule (i) gives (A_p) in length p. In rule (ii), the terms $x_\mu \cdot y$ and $[x_\lambda, x_\mu] \cdot z_T$ lie in S_p by (B_{p-1}). On S_{p-1}, both conditions were proved at stage $p-1$.

(C_p), the ordered case. Keep the notation of rule (ii). The monomial $z_\lambda z_T$ has length p, and $\mu < \lambda$, $\mu \le T$. So μ is at most every index of $z_\lambda z_T$, and rule (i) applies to $x_\mu \cdot (z_\lambda z_T)$:

$$x_\mu \cdot (x_\lambda \cdot z_T) = x_\mu \cdot (z_\lambda z_T) + x_\mu \cdot y = z_\mu z_\lambda z_T + x_\mu \cdot y = z_\lambda z_\Sigma + x_\mu \cdot y.$$

Also $x_\mu \cdot z_T = z_\Sigma$ by (A_{p-1}). Rule (ii) therefore gives

$$x_\lambda \cdot (x_\mu \cdot z_T) = x_\lambda \cdot z_\Sigma = z_\lambda z_\Sigma + x_\mu \cdot y + [x_\lambda, x_\mu] \cdot z_T,$$

that is,

$$x_\lambda \cdot (x_\mu \cdot z_T) = x_\mu \cdot (x_\lambda \cdot z_T) + [x_\lambda, x_\mu] \cdot z_T. \tag{11.1}$$

This is (C_p) for $z = z_T$ of length $p-1$ when $\mu < \lambda$ and $\mu \le T$. Exchanging λ and μ changes the sign of both sides of (C). So (C_p) also holds when $\lambda < \mu$ and $\lambda \le T$. It is trivial when $\lambda = \mu$. For $z \in S_{p-2}$ it is (C_{p-1}).

(C_p), the Jacobi case. Let z_T be an ordered monomial of length $p-1$. For $p = 1$, $T = \varnothing$ and an ordered case applies. For $p \ge 2$, let ν be the first index of T. If $\min(\lambda, \mu) \le \nu$, an ordered case applies. It remains to take $p \ge 2$ with $\nu < \lambda$ and $\nu < \mu$. Write $T = (\nu, \Psi)$. Since $\nu \le \Psi$, (A_{p-1}) gives

$$z_T = x_\nu \cdot z_\Psi, \qquad z_\Psi \in S_{p-2}.$$

On z_Ψ, (C_{p-1}) holds for all pairs of elements of $\mathfrak{g}$.

First, (C_{p-1}) gives

$$x_\mu \cdot z_T = x_\nu \cdot (x_\mu \cdot z_\Psi) + [x_\mu, x_\nu] \cdot z_\Psi.$$

Apply x_λ to both terms. For the second term, (C_{p-1}) gives

$$x_\lambda \cdot ([x_\mu, x_\nu] \cdot z_\Psi) = [x_\mu, x_\nu] \cdot (x_\lambda \cdot z_\Psi) + [x_\lambda, [x_\mu, x_\nu]] \cdot z_\Psi.$$

For the first term, use (B_{p-2}) to write

$$x_\mu \cdot z_\Psi = z_\mu z_\Psi + w, \qquad w \in S_{p-2}.$$

The monomial $z_\mu z_\Psi$ has length $p-1$, its indices are at least ν, and $\nu < \lambda$. So (11.1) applies to the pair λ, ν and this monomial. Also (C_{p-1}) applies to w:

$$x_\lambda \cdot (x_\nu \cdot (z_\mu z_\Psi)) = x_\nu \cdot (x_\lambda \cdot (z_\mu z_\Psi)) + [x_\lambda, x_\nu] \cdot (z_\mu z_\Psi),$$
$$x_\lambda \cdot (x_\nu \cdot w) = x_\nu \cdot (x_\lambda \cdot w) + [x_\lambda, x_\nu] \cdot w.$$

Adding the two lines,

$$x_\lambda \cdot (x_\nu \cdot (x_\mu \cdot z_\Psi)) = x_\nu \cdot (x_\lambda \cdot (x_\mu \cdot z_\Psi)) + [x_\lambda, x_\nu] \cdot (x_\mu \cdot z_\Psi).$$

Substituting both results,

$$x_\lambda \cdot (x_\mu \cdot z_T) = x_\nu \cdot (x_\lambda \cdot (x_\mu \cdot z_\Psi)) + [x_\lambda, x_\nu] \cdot (x_\mu \cdot z_\Psi) \\ + [x_\mu, x_\nu] \cdot (x_\lambda \cdot z_\Psi) + [x_\lambda, [x_\mu, x_\nu]] \cdot z_\Psi.$$

The hypotheses are symmetric in λ and μ. Exchange them and subtract. The two middle terms are symmetric and cancel:

$$\begin{aligned} & x_\lambda \cdot (x_\mu \cdot z_T) - x_\mu \cdot (x_\lambda \cdot z_T) \\ &\quad = x_\nu \cdot \big(x_\lambda \cdot (x_\mu \cdot z_\Psi) - x_\mu \cdot (x_\lambda \cdot z_\Psi)\big) + \big([x_\lambda, [x_\mu, x_\nu]] - [x_\mu, [x_\lambda, x_\nu]]\big) \cdot z_\Psi \\ &\quad = x_\nu \cdot ([x_\lambda, x_\mu] \cdot z_\Psi) + [[x_\lambda, x_\mu], x_\nu] \cdot z_\Psi \\ &\quad = [x_\lambda, x_\mu] \cdot (x_\nu \cdot z_\Psi) \\ &\quad = [x_\lambda, x_\mu] \cdot z_T. \end{aligned}$$

The second equality uses (C_{p-1}) and the Jacobi identity in the form

$$[x_\lambda, [x_\mu, x_\nu]] = [[x_\lambda, x_\mu], x_\nu] + [x_\mu, [x_\lambda, x_\nu]].$$

The third uses (C_{p-1}) for the pair $[x_\lambda, x_\mu]$ and x_ν. Linearity in z gives (C_p). ∎

Proof of Theorem 11.6. Spanning is Proposition 11.5. Define the map $\rho\colon \mathfrak{g} \to \mathrm{End}(S)$ by

$$\rho(x) z = x \cdot z.$$

By (C) and linearity, ρ is a representation. Let $\tilde{\rho}$ be its extension to $U(\mathfrak{g})$ from Proposition 11.3. We claim

$$\tilde{\rho}(x_\Sigma) 1 = z_\Sigma.$$

Use induction on the length. For $\Sigma = \varnothing$ both sides are 1. For $\Sigma = (\lambda, \Sigma')$ we have $\lambda \le \Sigma'$. Since $\tilde{\rho}$ is a homomorphism, the induction hypothesis and (A) give

$$\tilde{\rho}(x_\Sigma) 1 = \rho(x_\lambda) \tilde{\rho}(x_{\Sigma'}) 1 = x_\lambda \cdot z_{\Sigma'} = z_\lambda z_{\Sigma'} = z_\Sigma.$$

If $\sum c_\Sigma x_\Sigma = 0$, apply $\tilde{\rho}$ of this element to 1. Then $\sum c_\Sigma z_\Sigma = 0$. The monomials z_Σ are independent, so every c_Σ is zero. ∎

11.4 Consequences

Corollary 11.8. *The map ι is injective. The ordered monomials of length at most p form a basis of U_p. Hence $\dim U_p = \binom{n+p}{p}$, the number of monomials of degree at most p in n variables.*

Proof. The elements $\iota(x_i)$ are distinct ordered monomials of length one, so they are independent. The second statement combines Proposition 11.5 with Theorem 11.6. ∎

From now on $\mathfrak{g}$ is identified with $\iota(\mathfrak{g})$.

The *associated graded algebra* is

$$\operatorname{gr} U(\mathfrak{g}) = \bigoplus_{p \ge 0} U_p / U_{p-1}, \qquad (a + U_{p-1})(b + U_{q-1}) = ab + U_{p+q-1},$$

for $a \in U_p$ and $b \in U_q$. The product is well defined because $U_{p-1}U_q$ and $U_p U_{q-1}$ lie in U_{p+q-1}. The *symmetric algebra* $\mathrm{Sym}(\mathfrak{g})$ is $T(\mathfrak{g})$ modulo the ideal generated by all $x \otimes y - y \otimes x$. The chosen basis of $\mathfrak{g}$ identifies $\mathrm{Sym}(\mathfrak{g})$ with S, by $x_i \mapsto z_i$.

Theorem 11.9. *The map $z_i \mapsto x_i + U_0$ extends to an isomorphism of graded algebras*

$$\psi\colon \operatorname{Sym}(\mathfrak{g}) \longrightarrow \operatorname{gr} U(\mathfrak{g}), \qquad z_\Sigma \longmapsto x_\Sigma + U_{p-1} \quad (\Sigma \text{ of length } p).$$

Proof. For $x, y \in \mathfrak{g}$, the commutator of $x + U_0$ and $y + U_0$ in $\operatorname{gr} U(\mathfrak{g})$ is

$$[x, y] + U_1 = 0 \quad \text{in } U_2/U_1,$$

since $[x, y] \in U_1$. By Proposition 11.5, the classes $x_i + U_0$ generate $\operatorname{gr} U(\mathfrak{g})$. These generators commute, so $\operatorname{gr} U(\mathfrak{g})$ is commutative. The polynomial algebra S is free commutative on the z_i. Hence there is an algebra homomorphism ψ with the displayed values.

By Corollary 11.8, the classes of ordered monomials of length exactly p form a basis of U_p/U_{p-1}. Thus ψ maps the monomial basis of degree p bijectively onto a basis of U_p/U_{p-1}. ∎

Corollary 11.10 (Factorisation). *Let $\mathfrak{a}, \mathfrak{k} \subseteq \mathfrak{g}$ be subalgebras with $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{k}$ as vector spaces. The multiplication map*

$$U(\mathfrak{a}) \otimes U(\mathfrak{k}) \longrightarrow U(\mathfrak{g}), \qquad u \otimes v \longmapsto uv,$$

is a linear isomorphism. For any subalgebra $\mathfrak{a} \subseteq \mathfrak{g}$, the induced map $U(\mathfrak{a}) \to U(\mathfrak{g})$ is injective.

Proof. The universal property gives the maps $U(\mathfrak{a}) \to U(\mathfrak{g})$ and $U(\mathfrak{k}) \to U(\mathfrak{g})$ induced by inclusion. Order a basis of $\mathfrak{g}$ by listing a basis of $\mathfrak{a}$ first and then a basis of $\mathfrak{k}$. Each ordered monomial of $\mathfrak{g}$ is uniquely a product

$$x_\Sigma = x_{\Sigma'} x_{\Sigma''},$$

where $x_{\Sigma'}$ is an ordered monomial of $\mathfrak{a}$ and $x_{\Sigma''}$ one of $\mathfrak{k}$. By PBW for $\mathfrak{a}$ and $\mathfrak{k}$, the tensors $x_{\Sigma'} \otimes x_{\Sigma''}$ form a basis of $U(\mathfrak{a}) \otimes U(\mathfrak{k})$. The multiplication map sends this basis bijectively onto the PBW basis of $U(\mathfrak{g})$.

For any subalgebra, extend its ordered basis to an ordered basis of $\mathfrak{g}$. PBW shows that the induced map sends distinct basis monomials to distinct basis monomials. ∎

Now let $\mathfrak{g}$ be semisimple. Choose positive roots as in Lecture 9 and put

$$\mathfrak{n}_\pm = \bigoplus_{\alpha \in R_+} \mathfrak{g}_{\pm\alpha}.$$

The root decomposition gives $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$, the *triangular decomposition*. The rules $[\mathfrak{h}, \mathfrak{g}_\alpha] \subseteq \mathfrak{g}_\alpha$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$ of Lecture 7 show that $\mathfrak{n}_\pm$ and $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$ are subalgebras. Corollary 11.10, applied to $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{b}$ and to $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}_+$, gives

$$U(\mathfrak{n}_-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}_+) \xrightarrow{\cong} U(\mathfrak{g}), \qquad U(\mathfrak{n}_-) \otimes U(\mathfrak{b}) \xrightarrow{\cong} U(\mathfrak{g}).$$

The second isomorphism makes $U(\mathfrak{g})$ a free right $U(\mathfrak{b})$-module with basis the ordered monomials of $\mathfrak{n}_-$. Lecture 12 uses this to construct induced modules. Since $\mathfrak{h}$ is abelian, the defining relations of $U(\mathfrak{h})$ and $\operatorname{Sym}(\mathfrak{h})$ coincide, so $U(\mathfrak{h}) = \operatorname{Sym}(\mathfrak{h})$.

11.5 Reordering in $U(\mathfrak{sl}_2)$

Order the basis of $\mathfrak{sl}_2$ as (f, h, e), matching $\mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$. The ordered monomials are $f^a h^b e^c$. The brackets $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$ of Lecture 6 give the rewriting rules

$$eh = he - 2e, \qquad hf = fh - 2f, \qquad ef = fe + h.$$

Example 11.11 (The word ehf)**.** Move e to the right first:

$$\begin{aligned} ehf &= (he - 2e)f \\ &= h(ef) - 2ef \\ &= h(fe + h) - 2(fe + h) \\ &= (hf)e + h^2 - 2fe - 2h \\ &= (fh - 2f)e + h^2 - 2fe - 2h \\ &= fhe - 4fe + h^2 - 2h. \end{aligned}$$

PBW gives uniqueness of this ordered expansion.

Now compare with $\operatorname{gr} U(\mathfrak{sl}_2)$. Write z_f, z_h, z_e for the variables of $\operatorname{Sym}(\mathfrak{sl}_2)$. The word ehf lies in U_3, and the correction terms lie in U_2. In U_3/U_2,

$$ehf + U_2 = fhe + U_2 = \psi(z_f z_h z_e).$$

In the associated graded algebra, the reordering is plain commutativity. In $U(\mathfrak{sl}_2)$ itself,

$$ehf - fhe = -4fe + h^2 - 2h \neq 0$$

by PBW independence. The term $-2h$ lies in U_1, so

$$ehf - fhe + U_1 = \psi(-4 z_f z_e + z_h^2) \neq 0 \quad \text{in } U_2/U_1.$$

11.6 Coproduct and antipode

The algebra $U(\mathfrak{g}) \otimes U(\mathfrak{g})$ has product $(a \otimes b)(c \otimes d) = ac \otimes bd$. The *opposite algebra* $U(\mathfrak{g})^{\mathrm{op}}$ has product $a * b = ba$. An *antihomomorphism* is a linear map $S \colon U(\mathfrak{g}) \to U(\mathfrak{g})$ that satisfies $S(1) = 1$ and $S(uv) = S(v)S(u)$.

Proposition 11.12. *There is a unique algebra homomorphism $\Delta \colon U(\mathfrak{g}) \to U(\mathfrak{g}) \otimes U(\mathfrak{g})$ and a unique antihomomorphism $S \colon U(\mathfrak{g}) \to U(\mathfrak{g})$ such that for $x \in \mathfrak{g}$*

$$\Delta(x) = x \otimes 1 + 1 \otimes x, \qquad S(x) = -x.$$

Proof. Both statements follow from Proposition 11.2 once the bracket condition is checked. For Δ,

$$(x \otimes 1 + 1 \otimes x)(y \otimes 1 + 1 \otimes y) = xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy.$$

Exchange x and y and subtract. The two middle terms cancel:

$$[\Delta(x), \Delta(y)] = [x, y] \otimes 1 + 1 \otimes [x, y] = \Delta([x, y]).$$

For S, an antihomomorphism is a homomorphism into $U(\mathfrak{g})^{\mathrm{op}}$. Apply Proposition 11.2 to the map $\mathfrak{g} \to U(\mathfrak{g})^{\mathrm{op}}$, $x \mapsto -x$. Its bracket condition uses the opposite product:

$$(-x) * (-y) - (-y) * (-x) = (-y)(-x) - (-x)(-y) = yx - xy = -[x, y]. \qquad \blacksquare$$

Lecture 14 uses Δ and S to define tensor products and duals of modules.

Self-checks

(1) Write eh^2 in the basis $f^a h^b e^c$ of $U(\mathfrak{sl}_2)$.
(2) List a basis of $U_2 \subseteq U(\mathfrak{sl}_2)$ and state its dimension.
(3) Express $S(ef)$ in the basis $f^a h^b e^c$.

Answers. (1) Use $eh = (h-2)e$ twice:

$$eh^2 = (h-2)eh = (h-2)^2 e = h^2 e - 4he + 4e.$$

(2) The ordered monomials $1, f, h, e, f^2, fh, fe, h^2, he, e^2$ form a basis. The dimension is $\binom{5}{2} = 10$. (3) $S(ef) = S(f)S(e) = (-f)(-e) = fe$, already ordered.

Summary

The enveloping algebra $U(\mathfrak{g})$ turns brackets into commutators. Its unital modules are exactly the $\mathfrak{g}$-modules.

Reordering proves that the ordered monomials span $U(\mathfrak{g})$. Their independence comes from a polynomial module whose construction uses the Jacobi identity. Two consequences are $\operatorname{gr} U(\mathfrak{g}) \cong \operatorname{Sym}(\mathfrak{g})$ and, for semisimple $\mathfrak{g}$, $U(\mathfrak{g}) \cong U(\mathfrak{n}_-) \otimes U(\mathfrak{b})$ as vector spaces. The second is the input for Verma modules in Lecture 12.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 17.1–17.4; the independence proof follows Section 17.4. Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd edition, Chapter 9.